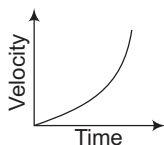


> PRACTICE EXERCISE

Measurement, Motion, Work, Energy and Power

1. A train travels a distance with a speed of 30 km/h and returns with a speed of 50 km/h. The average speed of the train is
(a) 36 km/h (b) 37.5 km/h
(c) 38 km/h (d) 40 km/h
2. Average velocity of an object is equal to the mean of its initial and final velocities, if the acceleration is
(a) uniform (b) variable
(c) Both a and b (d) None of these
3. A water tank filled upto $\frac{2}{3}$ of its height is moving with a uniform speed. On sudden application of the brake, the water in the tank would
(a) move backward (b) move forward
(c) be unaffected (d) rise upwards
4. When the distance, an object travels is directly proportional to the length of time, it is said to travel with
(a) zero velocity (b) uniform velocity
(c) constant velocity (d) constant acceleration
5. If the displacement of an object is proportional to square of time, then the object moves with
(a) uniform velocity
(b) uniform acceleration
(c) increasing acceleration
(d) decreasing acceleration
6. A passenger in a moving train tosses a five rupee coin. If the coin falls behind him, the train must be moving with a uniform
(a) acceleration
(b) deceleration
(c) speed
(d) velocity
7. The velocity-time graph for an object is shown below. Which one of the following statements holds true for this object?



- (a) The object is moving with uniform speed
(b) The object is in non-uniform accelerated
(c) The object is at rest
(d) The object is in non-uniform decelerated.
8. Which of the following figure represents uniform motion of a moving object correctly?

(a) Distance vs Time: straight line through origin
(b) Distance vs Time: curve starting from origin, concave up
(c) Distance vs Time: curve starting from origin, concave down
(d) Distance vs Time: horizontal line
9. A body falls from rest. The velocity acquired by falling a distance $2h$ is
(a) $2\sqrt{gh}$ (b) $\sqrt{gh}$
(c) $\sqrt{2gh}$ (d) None of these
10. The velocity-time graph for an object is shown. Which one of the following statements holds true for this object?

The graph shows Velocity on the vertical axis and Time on the horizontal axis. The line is horizontal, indicating constant velocity.

(a) The object has uniform velocity
(b) The object has uniform acceleration
(c) The object has non-uniform acceleration
(d) None of the above
11. A boy throws four stones of same shape, size and weight with equal speed at different initial angles with the horizontal line. If the angles are 15° , 30° , 45° and 60° , at which angle the stone will cover the maximum distance horizontally?
(a) 15° (b) 30° (c) 45° (d) 60°
12. Which one of the following is a vector quantity?
(a) Force (b) Area
(c) Volume (d) Density
13. External forces acting on an object
(a) are always balanced
(b) are always unbalanced
(c) may or may not be balanced
(d) None of the above
14. When horse starts running all of a sudden, the rider on the horse falls back because
(a) he is afraid
(b) he is taken a back
(c) due to inertia of motion, the lower part of his body comes in motion
(d) due to inertia of rest, the upper part of his body remains at rest
15. A scooter driven over an oily road generally slips because on the oily road
(a) inertia of the scooter tyres increases
(b) inertia of the scooter tyres decreases
(c) friction between tyres and road increases
(d) friction between tyres and road decreases
16. Newton's second law of motion connects
(a) momentum and acceleration
(b) change of momentum and velocity
(c) rate of change of momentum and external force
(d) rate of change of force and momentum
17. When a ball drops onto the floor it bounces. Why does it bounce?
(a) Newton's third law implies that for every action (drop) there is a reaction (bounce)
(b) The floor exerts a force on the ball during the impact
(c) The floor is perfectly rigid
(d) The floor heats up on impact
18. If an object having mass of 1 kg is subjected to a force of 1 N, it moves with
(a) a speed of 1 m/s
(b) a speed of 1 km/s
(c) an acceleration of 10 m/s^2
(d) an acceleration of 1 m/s^2
19. Momentum has the same unit as that of
(a) couple
(b) torque
(c) impulse
(d) moment of momentum

- 20.** Rocket works on the principle of conservation of

(a) mass (b) energy
(c) momentum (d) None of these

- 21.** Two balls of different masses have the same kinetic energy. The ball having greater momentum will be

(a) lighter one
(b) heavier one
(c) both having equal masses
(d) cannot say

- 22.** A goalkeeper in a game of football pulls his hands backwards after holding the ball shot at the goal. This enables the goalkeeper to

(a) exert larger force on the ball
(b) reduce the force exerted by the ball on hands
(c) increase the rate of change of momentum
(d) decrease the rate of change of momentum

- 23.** A cricket player moves his hands along the direction of the moving ball while catching it in order to

(a) kill the googly of the ball
(b) conserve energy
(c) conserve mass
(d) reduce the force of impulse

- 24.** Consider the following statements

1. The impulse given to a body is equal to the change in the momentum of the body.
2. The unit of the impulse is the same as that of the momentum.
3. The product of the force and velocity is called the impulse.
4. Impulse is a vector quantity.

Which of the following statements is/are correct?

(a) 1, 2 and 4 (b) Only 4
(c) Only 3 (d) 1, 2 and 3

- 25.** It is easier to roll a barrel along the road than to pull it because

(a) rolling involves only one point of contact between the barrel and the earth
(b) rolling friction is considerably less than the sliding friction
(c) the full weight of the barrel does not become effective while rolling
(d) None of the above

- 26.** It is difficult to walk on ice than on road because

(a) ice is harder than road
(b) road is harder than ice
(c) ice does not offer any reaction, when we push it, with our foot
(d) ice has a lesser friction than road

- 27.** Frictional force

1. is self-adjusting force.
2. is a non-conservative force.
3. is a necessary evil.

Codes

(a) 1 and 2 (b) 1 and 3
(c) Only 1 (d) All of these

- 28.** Consider the following statements

1. Friction force is conservative force.
2. Magnetic force and electrostatic force are non-conservative force.

Which of the following sentences is/are correct?

(a) Only 1 (b) Only 2
(c) 1 and 2 (d) None of these

- 29.** The chance of a vehicle to overturn while negotiating a circular path depends on

(a) speed of the vehicle only
(b) radius of the circular path only
(c) speed and height of the vehicle and radius of the circular path
(d) height of the vehicle only

- 30.** Fat can be separated from milk in a cream separation because of

(a) cohesive force
(b) gravitational force
(c) centrifugal force
(d) centripetal force

- 31.** Consider the following statements

1. If a body is rotating in a circular path, then after one rotation its distance will be zero.
2. distance and displacement are equal in straight line.
3. The speedometer of the car shows its instantaneous speed.

Which of the following statements is/are correct?

(a) Only 1 (b) Only 2
(c) 2 and 3 (d) 1 and 3

- 32.** Match the following columns.

Column I	Column II
A. We hit a carpet with a stick to remove the dust	1. Law of motion II
B. China wires are wrapped in straw or paper	2. Law of motion I
C. A jet plane moves	3. Law of motion III
D. Bogies of the train are provided with buffers	4. Impulse

Codes

A B C D A B C D
(a) 3 4 1 2 (b) 2 1 3 4
(c) 1 2 3 4 (d) 3 4 2 1

- 33.** Match the following columns.

Column I (Situation)	Column II (Centripetal force)
A. Revolution of the earth around the sun	1. Tension in the string
B. Vehicle taking a turn on level road	2. Friction force exerted by the road on the tyres
C. A vehicle on a speed braker	3. Weight of the body
D. A particle tied to a string and whirled in a horizontal circle	4. Gravitation force exerted by the sun

Codes

A B C D A B C D
(a) 4 2 3 1 (b) 2 3 4 1
(c) 4 3 2 1 (d) 1 2 3 4

- 34.** When work is done on the body

(a) it gains energy
(b) it loses energy
(c) its energy remains constant
(d) None of the above

- 35. Statement I** When the force retards the motion of a body, the work done is zero.

Statement II Work done depends on angle between force and displacement.

Codes

(a) Both the statements are correct and statement II is the correct explanation of statement I
(b) Both the statements are correct, but statement II is not the correct explanation of statement I
(c) Statement I is correct, but statement II is incorrect
(d) Statement I is incorrect, but statement II is correct

- 36.** An object of mass 2 kg is lifted vertically through a distance of 1.5 m. The work done in process is

(a) 29.4 J (b) 19.4 J
(c) 17.4 J (d) 20.4 J

- 37.** The work done on an object does not depends upon the

(a) displacement (b) force applied
(c) final velocity (d) initial velocity

- 38.** Consider the following statements

1. When a body falls freely under gravity the work done by gravity is positive.
2. If a force of 1 dyne produces a displacement of 1 cm in the direction of the force, the work done is called 1 erg.
3. Work is scalar quantity.
4. Work done is always positive.

Which of the following statements is/are correct?

- (a) 1, 2 and 4 (b) 2 and 3
(c) Only 4 (d) 1, 2 and 3

- 39.** If the velocity of a particle is reduced to half of its initial value, then the kinetic energy of the particle will
(a) get doubled
(b) become four times
(c) reduce to half of its original value
(d) reduce to one-fourth of its original value

- 40.** A horse and a dog are running with the same speed. Then, the ratio of their kinetic energies, if the weight of the horse is ten times that of dog, is
(a) 10 : 1 (b) 100 : 1
(c) 1 : 9 (d) 9.8 : 98

- 41.** If acceleration due to gravity is 10 m/s^2 , then the potential energy of a body of mass 1 kg kept at a height of 5 m, is
(a) 50 J (b) 100 J (c) 500 J (d) 50 KJ

- 42.** When a body falls freely
(a) its PE is converted into KE
(b) its KE is converted into PE
(c) its mechanical energy is converted into heat energy
(d) None of the above

- 43.** Consider the following statements
1. When a body falls its potential energy decrease but at the same time kinetic energy increases by an equal amount.
2. The total amount of energy in the universe remains constant.
Which of the following statements is/are correct?

- (a) 1 and 2 (b) Only 1
(c) Only 2 (d) None of these

- 44.** If acceleration due to gravity is 9.8 m/s^2 . A bag of wheat weighs, 200 kg. The height to which it should be raised, so that its potential energy may be 9800 J, is
(a) 5 m (b) 4 m (c) 3 m (d) 10 m

- 45.** Two bodies of equal mass move with the velocity of $2v$ and $3v$, respectively. Then, the ratio of their kinetic energies is
(a) 4 : 9 (b) 2 : 3 (c) 1 : 2 (d) 8 : 11

- 46.** Power of the moving body is stored in the form of
(a) work and time
(b) force and distance
(c) force and velocity
(d) distance and acceleration

> Previous Years' Questions

- 47. Statement I** Pulling a lawn roller is easier than pushing it.

Statement II Pushing increases the apparent weight and hence the force of friction. **2012 II**

- (a) Both the statements are correct and statement II is the correct explanation of statement I
(b) Both the statements are correct, but statement II is not the correct explanation of statement I
(c) Statement I is correct, but statement II is incorrect
(d) Statement I is incorrect, but statement II is correct

- 48.** A bus travels at a speed of 50 km/h to go from its origin of its destination at a distance of 300 km and travels at a speed of 60 km/h to return to the origin. What is the average speed of the bus? **2012 II**

- (a) 54.55 km/h (b) 55 km/h
(c) 55.55 km/h (d) 54 km/h

- 49.** Magnetic, electrostatic and gravitational forces came under the category of **2013 II**

- (a) non-contact forces
(b) contact forces
(c) frictional forces
(d) non-frictional forces

- 50.** You are asked to jog in a circular track of radius 35 m. Right one complete round on the circular track, your displacement and distance covered by you respectively **2014 I**

- (a) zero and 220 m (b) 220 m and zero
(c) zero and 110 m (d) 110 m and 220 m

- 51.** In cricket match, while catching a fast moving ball, a fielder in the ground gradually pulls his hands backwards with the moving ball to reduce the velocity to zero. The act represents **2014 I**

- (a) Newton's first law of motion
(b) Newton's second law of motion
(c) Newton's third law of motion
(d) Law of conservation of energy

- 52.** The distance-time graph for an object is shown below. Which one of the following statements holds true for this object? **2014 II**



- (a) The object is moving with uniform speed
(b) The object is at rest
(c) The object is having non-linear motion
(d) The object is moving with non-uniform speed

- 53.** An electron and a proton are circulating with same speed in circular paths of equal radius. Which one among the following will happen, if the mass of a proton is about 2,000 times that of an electron? **2015 I**

- (a) The centripetal force required by the electron is about 2,000 times more than that required by the proton
(b) The centripetal force required by the proton is about 2,000 times more than that required by the electron
(c) No centripetal force is required for any charged particle
(d) Equal centripetal force acts on both the particles as they rotate in the same circular path

- 54.** An object is raised to a height of 3 m from the ground. It is then allowed to fall on to a table 1 m high from ground level. In this context, which one among the following statements is correct? **2015 I**

- (a) Its potential energy decreases by two-thirds its original value of total energy
(b) Its potential energy decreases by one-third its original value of total energy
(c) Its kinetic energy increases by two-third, while potential energy increases by one-third
(d) Its kinetic energy increases by one-third, while potential energy decreases by one-third

- 55. Statement I** When a gun is fired it recoils i.e. it pushes back, with much less velocity than the velocity of the bullet.

Statement II Velocity of the recoiling gun is less because the gun is much heavier than the bullet. **2015 I**

- (a) Both the statements are correct and statement II is the correct explanation of statement I
(b) Both the statements are correct, but statement II is not the correct explanation of statement I
(c) Statement I is correct but statement II is incorrect
(d) Statement I is incorrect but statement II is correct

- 56.** A body is falling freely under the action of gravity alone in vacuum. Which one of the following remains constant during the fall?
 (a) Potential energy **2015 II**
 (b) Kinetic energy
 (c) Total linear momentum
 (d) Total mechanical energy

- 57.** Newton's laws of motion do not hold good for objects **2015 II**
 (a) at rest
 (b) moving slowly
 (c) moving with high velocity
 (d) moving with velocity comparable to velocity of light

- 58.** A brick is thrown vertically from an aircraft flying 2 km above the Earth. The brick will fall with a
 (a) constant speed **2015 II**
 (b) constant velocity
 (c) constant acceleration
 (d) constant speed for sometime, then with constant acceleration as it nears the Earth

- 59.** The rate of change of momentum of a body is equal to the resultant **2016 I**
 (a) energy (b) power
 (c) force (d) impulse

- 60.** The SI unit of mechanical energy is **2016 I**
 (a) Joule (b) Watt
 (c) Newton-second (d) Joule-second

Rotational Motion and Gravitation

- 61.** The centre of mass of a system
 (a) is always at its geometric centre
 (b) is always anywhere inside it
 (c) is always outside it
 (d) may be inside or outside it
- 62.** Assertion (A) To unscrew a rusted nut, we need a wrench with longer arm.
 Reason (R) Wrench with longer arm reduces the torque of the arm.
 (a) Both A and R are correct and R is the correct explanation of the A
 (b) Both A and R are correct but R is not the correct explanation of the A
 (c) A is correct but R is incorrect
 (d) A is incorrect but R is correct
- 63.** Consider the following statements
 1. Torque is equal to the product of moment of inertia and angular acceleration.

2. Torque is a vector quantity.
 3. The rate of change of angular acceleration is equal to the torque.
 4. The turning effect of a body is known as torque.
 Which of the following statements is/are correct?
 (a) 1 and 2 (b) Only 4
 (c) 3 and 4 (d) 1, 2 and 4

- 64.** Consider the following quantities.

I. Angular momentum

II. Kinetic energy

Which of the quantities given above will remain constant for a satellite in orbit?

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 65.** A boy suddenly comes and sits on a circular rotating table. What will remain conserved?
 (a) Angular momentum
 (b) Potential energy
 (c) Linear momentum
 (d) Kinetic energy

- 66.** A man is sitting on a revolving stool with his arm outstretched. When he suddenly pulls his arms inside, then
 (a) his angular velocity decreases
 (b) his moment of inertia decreases
 (c) his angular velocity remains constant
 (d) his angular momentum increases

- 67.** The cat can survive fall from a height much more than human or any other animal. It is because the cat
 (a) can immediately adjust itself to land on all four paws and bend the legs to absorb the impact of falling
 (b) has elastic bones
 (c) has thick and elastic skin
 (d) also gets injury equally with other animals, but has tremendous endurance, resistance and speedy recovery

- 68.** An athlete diving off high springboard can perform a variety of exercise in tube air before entering the water body, which one of the following parameters will remain constant during the fall?
 (a) The athlete's linear momentum
 (b) The athlete's angular momentum
 (c) The athlete's kinetic energy
 (d) The athlete's moment of inertia

- 69.** A sphere of mass 40 kg is attracted by a second sphere of mass 60 kg with a force equal to 4×10^{-5} N. If G is 6×10^{-11} N-m²/k², what is the distance between them?
 (a) 7 cm (b) 6 cm (c) 10 cm (d) 12 cm

- 70.** If suddenly the gravitational force of attraction between the earth and a satellite revolving around it becomes zero, then the satellite will
 (a) continue to move in this orbit with same velocity
 (b) move tangentially to the original orbit in the same velocity
 (c) become stationary in its orbit
 (d) move towards the earth

- 71.** The force of gravitation between two bodies does not depend on
 (a) their separation
 (b) the product of their masses
 (c) the sum of their masses
 (d) the gravitational constant

- 72.** If the distance between two objects is doubled, the force of attraction between them will
 (a) become $\frac{1}{4}$ times (b) become 2 times
 (c) become $\frac{1}{8}$ times (d) remains same

- 73.** Consider the following statements
 1. Gravitational force is always attractive in nature.
 2. Gravitational force is a central force.
 3. Tides are formed in oceans due to gravitational attraction between moon and ocean water.

Which of the following statements is/are correct?

- (a) 1 and 2 (b) 2 and 3
 (c) 1 and 3 (d) All of these
- 74.** The acceleration due to gravity
 (a) has the same value everywhere in space
 (b) has the same value everywhere on the earth
 (c) varies with the latitude on the earth
 (d) is greater on the moon due to its smaller diameter
- 75.** Two objects of different masses falling freely near the surface of the moon would
 (a) have same velocities at any instant
 (b) have different accelerations
 (c) experience forces of same magnitude
 (d) undergo a change in their media

- 76.** An apple falls from a tree because of gravitational attraction between the earth and apple. If F_1 is the magnitude of force exerted by the earth on the apple and F_2 is the magnitude of force exerted by apple on the earth, then

(a) F_1 is very much greater than F_2
 (b) F_2 is very much greater than F_1
 (c) F_1 is only a little greater than F_2
 (d) F_1 and F_2 are equal

- 77.** When an object is thrown up, the force of gravity

(a) is opposite to the direction of motion
 (b) is in the same direction as the direction of motion
 (c) decreases as it rises up
 (d) increases as it rises up

- 78.** Assertion (A) Moon's gravity is equivalent to one-eighth of the earth's gravity.

Reason (R) Due to less gravity, moon is unable to retain its atmosphere.

(a) Both A and R are correct and R is the correct explanation of A
 (b) Both A and R are correct, but R is not the correct explanation of A
 (c) A is correct, but R is incorrect
 (d) A is incorrect, but R is correct

- 79.** A body has a mass of 6 kg on the earth, when measured on the moon, its mass would be

(a) nearly 1 kg (b) less than 1 kg
 (c) less than 6 kg (d) 6 kg

- 80.** Match the following columns.

Column I	Column II
A. g is maximum	1. At the moon
B. g is minimum	2. At pole
C. g is zero	3. At the equator
D. g is 1/6 of original value	4. At centre of the earth

Codes

A B C D A B C D
 (a) 1 2 3 4 (b) 2 3 4 1
 (c) 4 3 2 1 (d) 4 2 3 1

- 81.** Consider the following statements
 1. Mass of a body is not constant it changes from place to place.
 2. Mass is measured by an ordinary equal arm balance.
 3. At the equator, the mass of a body is maximum.

Which of the following statements is/are correct?

(a) 1 and 2 (b) Only 2
 (c) 1 and 3 (d) All of these

- 82.** A line drawn from the sun to a planet, moving around it, sweeps over a fixed area in a given interval of time. This is according to

(a) Kepler's law (b) Ohm's law
 (c) Lenz's law (d) Newton's law

- 83.** The earth revolves around the sun in an elliptical orbit, its speed

(a) is greatest when it is farthest from the sun
 (b) is greatest when it is closest of the sun
 (c) remains the same at all points on the orbit
 (d) goes on decreasing continuously

- 84.** The square of time period of revolution of a planet round the sun is

(a) directly proportional to its average distance from the sun
 (b) directly proportional to the square of its average distance from the sun
 (c) directly proportional to the cube of its average distance from the sun
 (d) inversely proportional to its average distance from the sun

- 85.** Consider the following statements

1. Time period of geo-stationary satellite is 24 h.
2. Geo-stationary satellite revolves in equatorial plane in the direction of the earth's rotation i.e. from West to East
3. Geo-stationary satellite revolves around the earth at 800 km.
4. INSAT 2B is a polar satellite.

Which of the following statements is/are correct?

(a) 1 and 2 (b) 2 and 3
 (c) 3 and 4 (d) 2, 3 and 4

- 86.** Two satellites are revolving in the same circular orbit, their

1. masses are same
2. velocities are same

Codes

(a) 1 and 2
 (b) Only 2
 (c) Only 1
 (d) None of the above

- 87.** A communication relay satellite is able to telecast programmes continuously from one part of the world to another, because

(a) its period of revolution is greater than the period of rotation of the earth on its axis
 (b) its period of revolution is equal to the period of rotation of the earth on its axis

(c) its period of revolution is less than the period of rotation of the earth on its axis
 (d) its mass is less than the mass of the earth

- 88.** A missile is launched with a velocity less than the escape velocity. The sum of its kinetic and potential energy is

(a) positive (b) negative
 (c) zero
 (d) may be positive or negative depending upon its initial velocity

- 89.** The escape velocity of a projectile from the surface of the earth is approximately

(a) 11.3 km/s (b) 11.2 km/s
 (c) 11.5 km/s (d) 11.4 km/s

- 90.** The velocity with which a projectile must be fired so that it escapes the earth's gravitation does not depend on

(a) mass of the earth
 (b) mass of the projectile
 (c) radius of the projectile's orbit
 (d) gravitational constant

- 91.** The escape velocity of a body on the surface of the earth is 11.2 km/s. If the earth's mass increases to twice its present value and the radius of the earth become half the escape velocity would become

(a) 5.6 km/s
 (b) 11.2 km/s (remain unchanged)
 (c) 22.4 km/s (d) 44.8 km/s

- 92.** The escape velocity at the moon

1. is 2.38 km/s
2. is less than the average velocity of the gas molecules at the temperature of the moon

Codes

(a) Only 1 (b) Only 2
 (c) 1 and 2
 (d) None of the above

- 93.** Match the following columns.

Column I (Quantity)	Column II (Definition)
A. Mass	1. Attractive force only
B. Gravitational force	2. The attractive force exerted by the earth
C. Escape velocity	3. It is a quantity of matter
D. Weight	4. Body must be projected up

Codes

A B C D A B C D
 (a) 2 1 3 4 (b) 3 1 4 2
 (c) 1 2 3 4 (d) 2 1 3 4